ic_autolab round 15 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T05:13:12Z at git `7d336d597f38cd97fc0015a46a4bf3d66c66ccbe`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-836de4` (`python3 tools/pcc/pcc.py standing PCC-20261104-836de4`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
28 **null**	ksum4f-n15	0.964 [0.927, 1.002]	1.022 [0.981, 1.065]	1.340 [1.274, 1.409]	unresolved inside the band	7.294 / 8.232 / 16.670	7.564 / 8.052 / 12.445	1.117 / 1.222 / 2.170	6.5x / 6.7x / 7.7x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

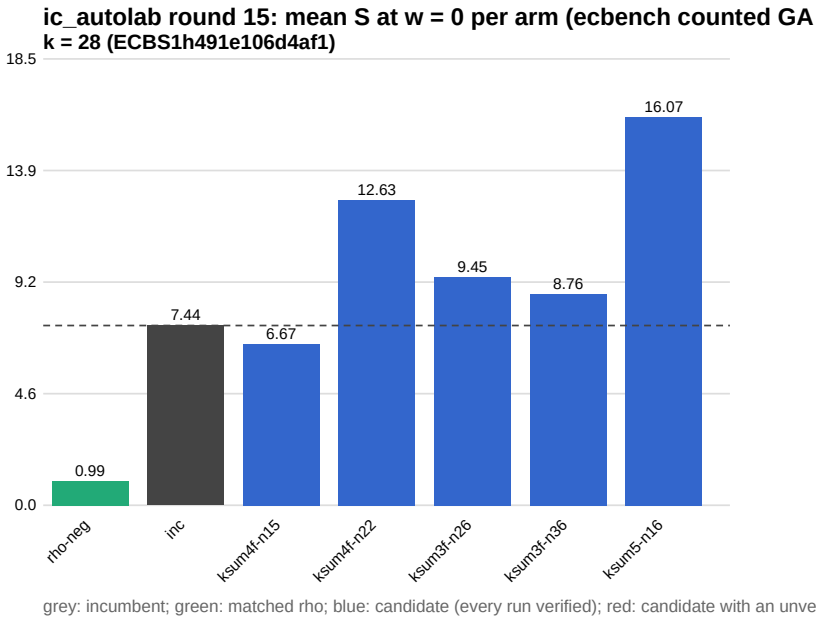
- k = 28: compare ops.status = ok, admissible = true; sel / inc = 0.9642 [0.927, 1.002] at w = 0; band: unresolved inside the band; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 28: session `ECBS1h491e106d4af1` (run exit 0, verify exit 0, receipt sha256 `e433b3fd2e73ef3fa781c3c6c0f465bf0d0b88e200fbb7bb2566ccf05bcb110f`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	0.993 / 1.085 / 1.921	matched reference
`inc`	ksum4	18 `prime-projected-abscissa:size=18,scan_cap=288`	`ksum:m=4`	32	32	7.440 / 7.906 / 12.101	incumbent
`ksum4f-n15`	ksum4f	15 `prime-projected-abscissa:size=15,scan_cap=240`	`ksum:m=4,free=1`	32	32	6.675 / 7.494 / 14.873	alternative ksum4f (best S(0) 7.330 at N = 18): N = 0.85 x 18
`ksum4f-n22`	ksum4f	22 `prime-projected-abscissa:size=22,scan_cap=352`	`ksum:m=4,free=1`	32	32	12.635 / 12.873 / 15.016	alternative ksum4f (best S(0) 7.330 at N = 18): N = 1.2 x 18
`ksum3f-n26`	ksum3f	26 `prime-projected-abscissa:size=26,scan_cap=416`	`ksum:m=3,free=1`	32	32	9.449 / 10.865 / 23.607	alternative ksum3f (best S(0) 8.208 at N = 30): N = 0.85 x 30
`ksum3f-n36`	ksum3f	36 `prime-projected-abscissa:size=36,scan_cap=576`	`ksum:m=3,free=1`	32	32	8.756 / 9.416 / 15.359	alternative ksum3f (best S(0)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
ksum5-n16	ksum5	16	prime-projected-abscissa:size=16,scan_cap=256	ksum:m=5	32	32	16.072 / 16.239 / 17.738	8.208 at N = 30): N = 1.2 x 30 exploration: family ksum5 untried at k = 28, N = 0.08 x N_a



train screen

Selection

- k = 28: challenger `ksum4f-n15` (Pareto alternative at w = 1: `ksum4f-n15`)

Holdout sessions

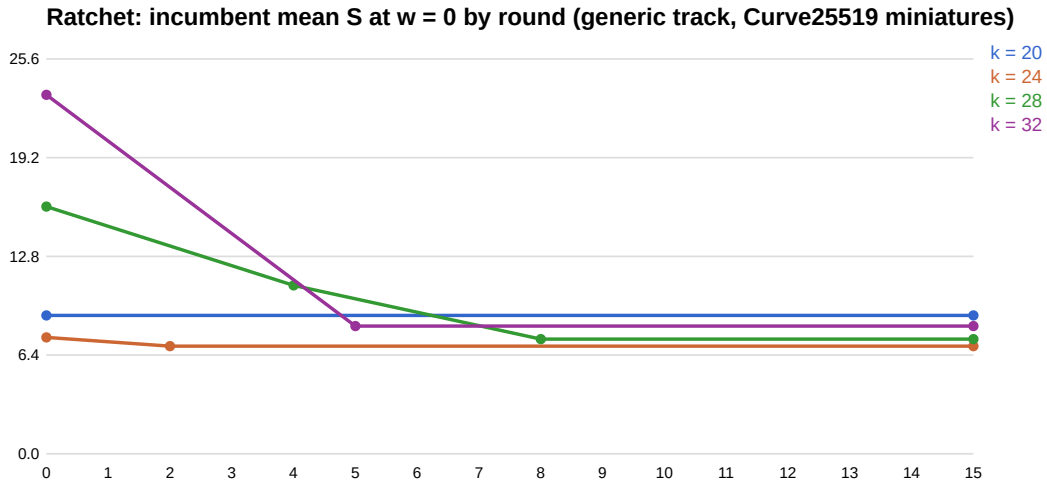
k	session	run exit	verify exit	receipt sha256	levels
28	ECBS1hb67f2de61d76	0	0	39c97f11518bc6707cadd17aeb36f1a30677096f82a7ecf7f1c53b76cef5fe2c	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	ksum3-n11 (prime-projected-abscissa:size=11,scan_cap=176 + ksum:m=3)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	ksum4-n12 (prime-projected-abscissa:size=12,scan_cap=192 + ksum:m=4)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x
28	ksum4-n18 (prime-projected-abscissa:size=18,scan_cap=288 + ksum:m=4)	8	7.445 / 7.910 / 12.100	7.8x / 7.6x / 6.5x
32	ksum4-n26 (prime-projected-abscissa:size=26,scan_cap=416	5	8.296 / 8.811 / 13.446	9.1x / 8.8x / 7.4x

k incumbent	since round S (w = 0 / 0.1 / 1)	distance to rho

+ `ksum:m=4`)



ratchet

Evidence and deviations

- ecbench binary sha256: `2dedee1fa0e91ee8161c47f825ff60dfaa00e4923cd6b2bb87afdd21bf238b4e`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.